

VIWA, Fiji

WMO: 916700

Lat: 17.150S

Lon: 176.900E

Elev: 16

StdP: 14.69

Time Zone: 12.00 (PCF)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	70.2	72.1	58.6	73.4	73.8	60.6	79.2	75.0	29.2	75.1	26.1	74.8	6.2	80	0.373

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	8.3	90.5	80.5	89.3	80.0	88.4	79.6	82.7	88.0	81.9	87.0	81.3	86.3	4.8	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
81.0	161.3	86.3	80.4	158.0	85.5	79.8	154.8	84.8	46.6	88.3	45.8	87.4	45.1	86.6	86.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.9	20.1	17.6		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			DB												
			WB												

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	80.6	82.7	83.0	82.8	81.7	80.0	79.5	77.9	77.8	78.9	79.9	81.2	82.4
	DBStd	2.82	2.15	1.97	1.90	2.31	3.37	2.07	1.79	1.94	2.00	1.98	1.95	1.96
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	2	0	0	0	0	2	0	0	0	0	0	0	0
	CDD50	11181	1015	924	1016	951	930	885	866	861	867	927	935	1004
	CDD65	5707	550	504	551	501	466	435	401	396	417	462	485	539
	CDH74	54789	6307	5887	6269	5292	4387	3581	2631	2512	3149	3988	4828	5958
	CDH80	14241	2162	2077	2078	1511	878	556	277	258	518	769	1260	1897
(14) Wind	WSAvg	6.9	6.1	5.6	5.4	5.5	6.3	8.1	8.8	9.3	7.8	6.7	6.0	6.6
(15) Precipitation	PrecAvg	74.6	9.3	7.8	8.7	8.6	6.9	4.7	3.2	2.8	4.1	5.7	5.6	6.9
	PrecMax	107.6	21.8	14.7	17.0	25.2	19.0	11.8	6.8	7.6	11.4	12.8	15.1	16.9
	PrecMin	51.2	0.6	1.0	2.1	0.4	0.9	0.7	0.5	0.4	0.2	1.1	1.1	0.6
	PrecStd	14.2	5.2	3.7	3.7	4.6	4.4	2.8	1.8	1.7	2.3	3.3	3.8	3.3
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	92.3	91.6	91.1	89.8	88.1	87.4	86.0	86.0	87.6	88.9	90.8	90.8
		MCWB	81.9	81.1	81.1	81.0	79.0	78.3	76.3	77.1	77.2	77.2	78.7	80.1
	2%	DB	90.2	89.8	89.1	88.4	86.6	85.7	84.1	84.0	85.7	86.7	88.1	89.1
		MCWB	80.7	80.7	80.5	79.8	78.1	77.2	75.2	76.0	76.1	76.8	77.7	79.4
	5%	DB	88.8	88.7	88.1	87.3	85.2	84.4	82.8	82.7	84.4	85.2	86.7	88.0
		MCWB	80.0	80.2	80.1	79.1	77.0	76.4	74.3	74.9	75.4	76.3	77.8	79.1
	10%	DB	87.6	87.4	87.1	86.2	84.1	83.2	81.6	81.5	83.0	83.8	85.5	86.9
		MCWB	79.6	79.7	79.7	78.7	76.3	75.8	73.6	74.1	74.8	75.9	77.6	78.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	83.5	83.7	83.4	82.8	80.8	80.7	78.9	79.2	79.5	79.7	81.5	82.8
		MCDB	89.9	88.6	88.4	88.6	85.5	85.0	83.0	83.4	84.6	84.9	87.0	87.8
	2%	WB	82.3	82.5	82.2	81.2	79.8	79.0	77.3	77.7	78.2	78.6	80.5	81.6
		MCDB	87.7	87.7	87.2	86.3	84.2	83.2	81.2	81.8	82.8	84.3	85.4	86.7
	5%	WB	81.5	81.6	81.4	80.3	78.9	78.1	76.2	76.5	77.2	77.8	79.6	80.7
		MCDB	86.7	86.6	86.3	85.1	83.3	82.2	80.5	80.7	81.6	83.0	84.4	85.9
	10%	WB	80.7	80.9	80.6	79.4	78.0	77.2	75.3	75.6	76.2	76.9	78.8	79.8
		MCDB	85.8	85.6	85.4	84.3	82.4	81.3	80.0	79.7	81.0	81.8	83.5	84.9
(35) Mean Daily Temperature Range	5% DB	MDBR	8.5	8.3	8.4	8.5	8.0	7.2	7.1	7.1	8.1	8.0	8.6	8.4
		MCDBR	10.9	10.2	9.7	10.0	9.5	8.7	8.8	8.8	10.0	10.0	10.4	10.1
	5% WB	MCWBR	4.9	4.7	4.7	4.9	4.7	4.2	4.2	4.1	4.6	4.4	4.6	4.6
		MCWBR	9.4	9.3	9.1	9.0	8.1	7.9	8.2	8.2	9.0	8.9	8.9	8.8
(40) Clear-Sky Solar Irradiance	taub		0.393	0.386	0.375	0.375	0.370	0.369	0.365	0.365	0.375	0.380	0.382	0.380
		taud	2.608	2.638	2.690	2.677	2.659	2.646	2.642	2.645	2.613	2.616	2.629	2.650
	Ebn at Noon		302	301	298	286	277	271	276	285	293	299	303	305
		Edn at Noon	33	32	29	28	26	26	27	28	31	32	32	31
(44) All-Sky Solar Radiation	RadAvg		2010	1913	1767	1553	1374	1285	1352	1517	1746	1886	1992	2033
	RadStd		211	165	137	116	113	81	70	140	134	134	194	108

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	+0.36	+0.41	N/A	N/A	+108	+108

Nomenclature: See separate page